

ANASTASIOS N. BATATEGAS

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EDUCATION

National Technical University of Athens (NTUA) Diploma (Integrated Master's) in Applied Mathematical and Physical Sciences Direction: Applied Physicist Specializations: Theoretical & Computational Physics, Nuclear Physics & Elementary Particles <i>Diploma Thesis (In Progress): To be announced. — Supervisor: Prof. K.N.A.</i>	<i>October 2021 - Present</i> Current Grade: 9.45/10.00
National Technical University of Athens (NTUA) School of Electrical & Computer Engineering (ECE) <i>Completed first-year coursework. Voluntarily transferred to the School of Applied Mathematical and Physical Sciences to strictly pursue fundamental and mathematical physics.</i>	<i>October 2020 - September 2021</i>
3rd General Lyceum of Trikala "Odysseas Elytis" High School Diploma	<i>September 2017 - June 2020</i> Grade: 20.0/20.0 (Valedictorian)

ACADEMIC PROJECTS & TERM PAPERS

SU(5) Grand Unified Theory <i>Course: Theoretical Physics</i>	<i>Fall 2026</i>
KATRIN Experiment: Direct Neutrino Mass Measurement <i>Course: Detector & Accelerator Technology</i> - Modeled the kinematics of molecular tritium β-decay and theoretical spectrum formatting near the 18.6 keV endpoint energy to extract the effective electron antineutrino mass. - Analyzed the electromagnetic mechanics and sub-eV energy resolution of MAC-E filters, alongside advanced background reduction techniques (e.g., Penning traps, Rydberg atoms).	<i>Fall 2025</i>
Bateman Equations in Radioactive Series <i>Course: Nuclear Physics & Applications</i> - Derived analytical solutions to the Bateman equations for generalized radioactive decay chains utilizing Laplace transforms and inductive mathematical proofs. - Investigated transient/secular equilibria and evaluated algorithmic solutions (e.g., matrix exponentials, divided differences) to mitigate numerical instabilities in complex decay networks.	<i>Spring 2025</i>
The Sudbury Neutrino Observatory (SNO): Oscillations & Analysis <i>Course: Detector & Accelerator Technology</i> - Authored a comprehensive analysis on the resolution of the Solar Neutrino Problem, mathematically detailing PMNS flavor oscillations and the Mikheyev-Smirnov-Wolfenstein (MSW) effect in matter. - Evaluated the statistical separation of Charged-Current, Neutral-Current, and Elastic Scattering reaction cross-sections in heavy water (D₂O) Cherenkov detectors.	<i>Fall 2024</i>

HONORS & AWARDS

SEEMOUS (South Eastern European Mathematical Olympiad for University Students) <i>Bronze Medal. Qualified during the Fall 2022 exams to represent NTUA and the Greek National Team in Struga, North Macedonia.</i>	<i>March 2023</i>
"Archimedes" Greek National Mathematical Olympiad <i>Silver Medal. Successfully passed the qualifying examinations to secure a position as a Substitute Member of the Greek National Team for the 2020 International Mathematical Olympiad (IMO).</i>	<i>2020</i>
Greek National Physics Competition <i>2nd Prize of Excellence.</i>	<i>2019</i>

TECHNICAL SKILLS & LANGUAGES

Programming: C/C++, Python, MATLAB, Fortran
Operating Systems & Tools: Linux/Unix, L^AT_EX
Languages: Greek (Native), English (Bilingual Proficiency – Cambridge C2 Proficiency & C1 Advanced)